

 Live Web Application: <https://roamcraft.onrender.com>
 GitHub Repository: <https://github.com/sagun-glitch/RoamCraft>

1. Executive Summary

RoamCraft is designed as a digital expedition dossier and field cartographer's trail map. Built with Node.js, Express, and Google Gemini, it streams personalized day-by-day itineraries live via Server-Sent Events (SSE) rendered across a topographic contour-line canvas with live compass bearing calculations.

2. Design Metaphor & Tokens

- Field Permit Registration Slip: Left permit panel styled on parchment (#E8DCC0) with stamped toggle chips and handwritten field margin notes.
- Topographic Map Canvas: Deep pine (#1B2A22) page background with SVG contour hairline rules (#C9B98C).
- Animated Route Line & Compass Rose: Copper trail path (#B5622E) that animates across the map as itineraries compile, paired with a rotating live compass needle.
- Typography: Header typography in Space Grotesk , data/body in IBM Plex Mono , and hand-lettered margin notes in Caveat .

3. Technical Stack & Architecture

Layer	Technology	Details
Live Hosting	Render Cloud Platform	Deployed live at https://roamcraft.onrender.com
Backend Engine	Node.js / Express.js	Manages REST endpoints, static file serving, and SSE streaming pipeline

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Generative AI	Google Gemini API	High-speed generative AI model configured for structured streaming output
Frontend UI	HTML5 / Vanilla CSS / ES6+ JS	Field Expedition Dossier built with Space Grotesk, IBM Plex Mono & Caveat fonts, and Lucide Icons
Streaming Protocol	Server-Sent Events (SSE)	HTTP streaming protocol for progressive chunk rendering in client browser

4. API Endpoint Reference

POST `/api/generate-itinerary`

Request Payload (JSON):

```
{
  "destination": "Kyoto, Japan",
  "days": 3,
  "budget": "Moderate",
  "interests": "temples, tea houses, ramen"
}
```

5. Metadata & Credits

- Project Name: RoamCraft
- Author / Owner: Sagun Mardi (`sagun-glitch`)
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